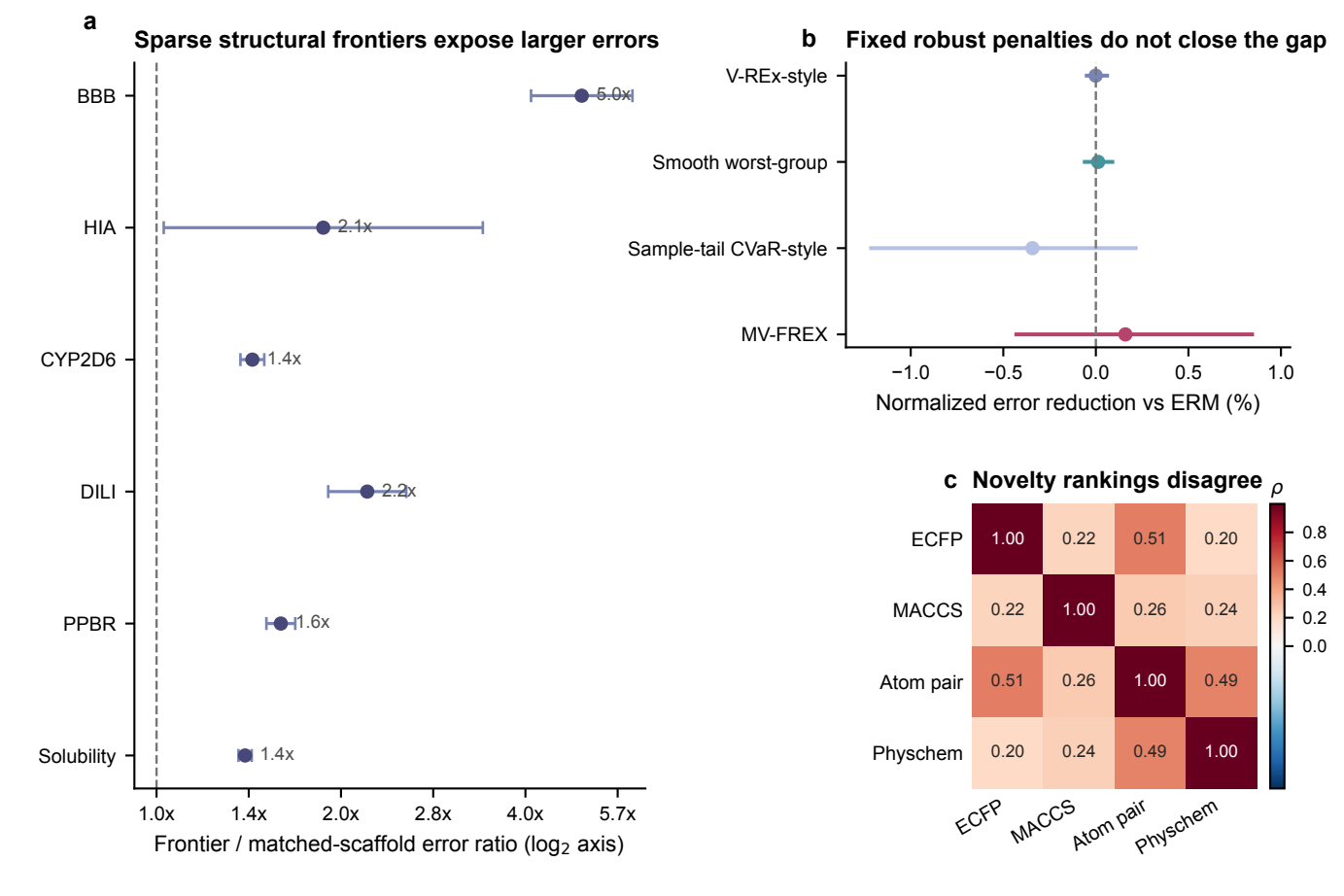
a
Sparse structural frontiers expose larger errors
BBB
HIA
CYP2D6
DILI
PPBR
Solubility
5.0x
2.1x
1.4x
2.2x
1.6x
1.4x
1.0x
1.4x
2.0x
2.8x
4.0x
5.7x
Frontier / matched-scaffold error ratio (log2 axis)
b
Fixed robust penalties do not close the gap
V-REx-style
Smooth worst-group
Sample-tail CVaR-style
MV-FREX
−1.0
−0.5
0.0
0.5
1.0
Normalized error reduction vs ERM (%)
c
Novelty rankings disagree
ρ
ECFP
MACCS
Atom pair
Physchem
1.00
0.22
0.51
0.20
0.22
1.00
0.26
0.24
0.51
0.26
1.00
0.49
0.20
0.24
0.49
1.00
0.8
0.6
0.4
0.2
0.0